Arun Rao Nayineni

Software Engineer | Distributed Systems | AI/ML Infrastructure | Agentic AI

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Education

Stevens Institute of Technology, M.S. in Computer Software Engineering — Hoboken, NJ
JNTUH, B.S. in Mechanical Engineering — Hyderabad, India

Technical Skills

Languages & Backend: Java, Python, C++, TypeScript, JavaScript, SQL, Spring Boot, FastAPI, REST APIs, microservices, multithreading
Distributed Systems & Cloud: AWS, Kafka, SQS, DynamoDB, S3, Redis, PostgreSQL, OpenSearch, Temporal, Docker, Kubernetes, Terraform, Jenkins, CI/CD
AI/ML & Agents: LangChain, LangGraph, StateGraph, MCP, multi-agent systems, RAG, tool calling, agent memory, human-in-the-loop, Hugging Face, PEFT, LoRA, RLHF
LLM Evaluation: Scenario replay, deterministic graders, LLM-as-Judge, groundedness, tool-trace validation, SQL safety, latency and token-cost analysis

Experience

Software Engineer, Amazon — **May 2025 – Present**

- Led a **zero-downtime cross-region migration** of a five-service, stateful forecasting platform spanning **7+ AWS data stores and 17 service dependencies**; designed shadow-traffic validation, phased rollout, and rollback strategies.
- Architected a **three-role LangGraph agent system** with a supervisor/router and two specialized agents, using typed StateGraph state, conditional routing, LangChain tools, and approval boundaries for internal AI-assisted workflows.
- Built **two-tier agent memory** with Redis checkpoints and PostgreSQL durable context, scoped access controls, and an evaluation harness for scenario replay, tool/schema/SQL validation, groundedness, and **three-provider** benchmarking.
- Built **MCP-based engineering automation** with three AI agents, nine tool integrations, and Temporal-backed orchestration, saving approximately **four weeks of engineering effort per service** during AWS migrations.
- Engineered a Java and AWS messaging-based real-time sorting system, improving outbound trailer SLA adherence by **87%**.
- Diagnosed cross-region authentication failures by decompiling Java bytecode and tracing AWS SigV4 flows; identified STS token incompatibility and delivered a system-wide fix.

Software Engineer, HCL Technologies — **May 2023 – Apr 2025**

- Developed a **human-feedback LLM evaluation pipeline** using Python and Hugging Face, transforming reviewer preferences into ranked response data and quality signals for relevance, factuality, and policy adherence.
- Productionized **LLM-backed services** using FastAPI, PostgreSQL, Redis, Docker, Terraform, Jenkins, and AWS, automating deployment and monitoring workflows.
- Designed **trade reconciliation workflows** to compare internal trade records with external counterparty data, detect discrepancies, and support exception resolution across post-trade operations.
- Built **Kafka-based event-driven pipelines** and integrated Elasticsearch for transaction state tracking, real-time processing, and operational visibility across middle and back-office systems.
- Developed **RESTful APIs** for payment processing and transaction management using TypeScript, Node.js, SQL, AWS Lambda, and Stripe.
- Optimized SQL queries, indexing, and caching to improve data retrieval performance by **30%**; reduced critical production defects by **70%**.

Software Engineer – Backend, Tata Consultancy Services — **Jan 2020 – Jul 2022**

- Developed REST APIs and **500+ unit tests**, contributing to a **40%** faster deployment cycle.
- Built Amazon **S3-to-Redshift ETL** pipelines for data cleansing, transformation, and loading.
- Automated CI/CD with Jenkins and Docker, reducing deployment latency by **70%** and anomalies by **90%**.

Junior Software Engineer, TalentSprint — **Jan 2019 – Dec 2019**

- Built Java/Spring Boot REST APIs supporting **10K+ concurrent users**, with Redis caching, JWT authentication, and multithreaded processing.
- Created database views and stored procedures and Integerated with Java/Spring Boot APIs to improve data retrieval efficiency